Academic Task Number: 2

Course code: CSE494

Date of allotment: 31st August, 2026

Course title: INTELLIGENT NOSQL DATABASES

Date of submission: 19th October, 2026

Maximum Marks: 30

Academic Task Type: Project

Question Number	Question Statement	Course Outcome	Bloom's level	Marks per Question
Q1	Design and implement a MongoDB-based application for a real-world problem of your choice. The project should demonstrate your ability to design a suitable document structure, apply validation rules, perform CRUD operations, connect MongoDB with Python using PyMongo, and retrieve meaningful information using aggregation queries. **Project Demonstration** During the presentation and viva, students should be able to explain: • The problem selected and its real-world relevance. • Database and document design. • Validation rules implemented. • CRUD queries used. • Aggregation pipelines used. • PyMongo implementation.	CO4, CO5, CO6	L2,L3	30

- Output obtained from the queries.
- Why particular queries or database structures were selected.

Deliverables

Students must submit:

- Project implementation containing MongoDB queries and PyMongo code.
- Project report explaining the problem, database design, validation, CRUD operations, aggregation queries, and PyMongo implementation.
- Presentation/demo of the working project.
- Screenshots or outputs demonstrating successful execution of the major operations.

Marks Distribution

Component	**Marks**
Project Implementation	15
Project Report	5
Presentation & Viva	10
Total	**30**

Important Instructions

- The project must be completed individually.
- The project must use MongoDB as the database.

	• PyMongo must be used for Python–MongoDB integration. • The project must include validation, CRUD operations, and aggregation. • Students should use their own dataset/scenario and avoid simply copying an existing project. • During the viva, students may be asked to modify or write queries based on their project data. • The submitted implementation should be executable and demonstrable.			